

Repeated Exam MA-224

Course code:	MA-224
Course name:	Discrete mathematics 2
Date:	20. February 2026
Time:	09:00 – 13:00 (4 hours)
Number of pages:	5
Number of tasks:	12
Permitted equipment:	All written or printed material. Calculator without scripts. No communication is allowed during the exam.
Remarks:	Present your answers so that it is clear what methods you have used, and how you have used them. Your answer should contain a calculation or an argument explaining how the answer was reached; otherwise, it is incomplete. All problems 1-12 are scored equally. If a problem has more than one part, then each part is scored equally within the problem.

Problem 1

We consider the following theorem.

$$\text{Th1: } [(p \vee q) \rightarrow r] \iff [(q \rightarrow r) \wedge (p \rightarrow r)]$$

- (a) Prove Th1 using truth tables.
- (b) Prove Th1 using logical inferences.

For your convenience, here are some possible inference rules.

equivalence between $\rightarrow$ and $\wedge$: $A \rightarrow B \iff \neg A \vee B$

deMorgan's Law for $\wedge$: $\neg(A \wedge B) \iff \neg A \vee \neg B$

deMorgan's Law for $\vee$: $\neg(A \vee B) \iff \neg A \wedge \neg B$

commutativity of $\wedge$: $A \wedge B \iff B \wedge A$

associativity of $\wedge$: $(A \wedge B) \wedge C \iff A \wedge (B \wedge C)$

distributivity of $\wedge$: $(A \vee B) \wedge C \iff (A \wedge C) \vee (B \wedge C)$

commutativity of $\vee$: $A \vee B \iff B \vee A$

associativity of $\vee$: $(A \vee B) \vee C \iff A \vee (B \vee C)$

distributivity of $\vee$: $(A \wedge B) \vee C \iff (A \vee C) \wedge (B \vee C)$

Problem 2

Define the following sequence recursively: $s_i = 3 \cdot (i + 4)$.

Problem 3

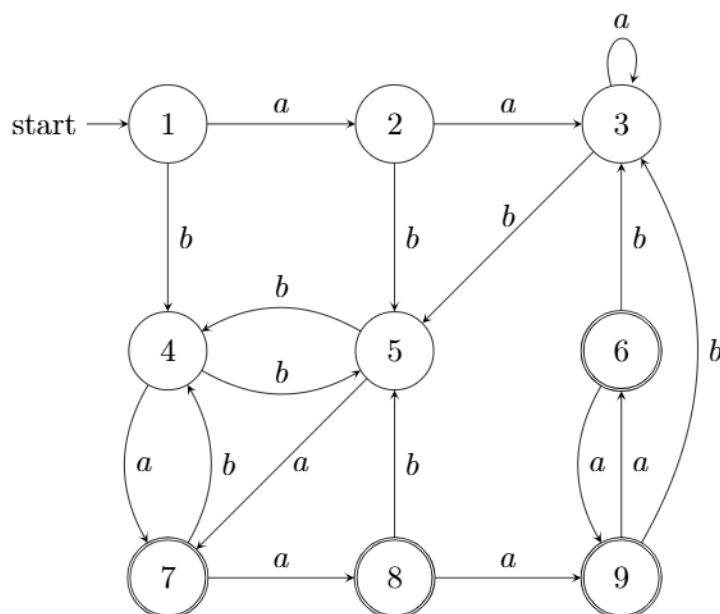
Consider the following EBNF grammar with the start symbol s . Terminals are enclosed in single quotes ('inc'). Alternatives are denoted by '|'.

$$s \rightarrow \text{'inc'} \mid \text{'dec'} \mid \text{'repeat' } s \text{'until' } s \mid s \text{'after' } s$$

- (a) Find a derivation in the EBNF grammar above for the string
repeat dec until inc after dec .
- (b) Is the grammar ambiguous? Give reasons.

Problem 4

Minimize the following DFA. Show all relevant steps.



Problem 5

Is the code below correct with respect to the following specification?

Specification: $\text{DIV}(a,b,\text{res}:\text{NUM}) \text{ res} := \text{res}' \mid a * \text{res}' = b$

Code: `float sqrt(a,b:float) { return b/a; }`

Problem 6

Let R be a symmetric and transitive relation of natural numbers. Each natural number is related with R to at least one natural number.

Prove that R is an equivalence relation or provide an example where this is not the case.

Problem 7

Find the solution to the initial value problem

$$\begin{cases} a_n - 4a_{n-1} - 4a_{n-2} = 0 \\ a_0 = 1 \\ a_1 = 2. \end{cases}$$

Problem 8

Assume that G is a group of order 8 and that its elements are named 1, a, b, c, d, e, f and g . The following is the multiplication table for G where a few entries are missing.

	1	a	b	c	d	e	f	g
1	1	a	b	c	d	e	f	g
a	a	b	d	g	1	c	e	f
b	b	d			a	g	c	e
c	c	g	f	b	e	a		d
d	d	1	a		b	f	g	c
e	e	c	g	a	f	1	d	b
f	f	e	c	1	g	d		a
g	g	f	e	d	c	b	a	1

(a) Compute the missing entries in the multiplication table:

$$b \cdot b =$$

$$b \cdot c =$$

$$c \cdot f =$$

$$d \cdot c =$$

$$f \cdot f =$$

(b) Prove or disprove that G is an Abelian group.

(c) Prove or disprove that G is a cyclic group.

Problem 9

Let $\phi(n) = |P_n|$ be the Euler totient function.

(a) Find a natural number n such that $\phi(n) > \phi(n+1)$.

(b) Use the definition of $\phi(n)$ to prove that $\phi(2^k) = 2^{k-1}$, for every positive integer $k \geq 1$.

Problem 10

Let G be a group of order 55 and let $H \subset G$ be a proper subgroup, i.e. that $\{1\} \neq H \neq G$. Use Lagrange's theorem to prove that the index $[G : H]$ is a prime number.

Problem 11

Let the encoding function

$$E : (\mathbb{Z}/2)^{\times 3} \rightarrow (\mathbb{Z}/2)^{\times 5}$$

be given by multiplication with the matrix:

$$E(w_1 \ w_2 \ w_3) = (w_1 \ w_2 \ w_3) \cdot \begin{bmatrix} 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 \end{bmatrix}$$

The message x is encoded using E and transmitted over a noisy channel. The bitstring $r = (11001)$ is received. We assume that the transmission error weight is at most 1 during this transmission.

- (a) Prove that r is a code word and find the original message x .
- (b) Prove that E is able to detect all transmission errors of weight 1.

Problem 12

Given the (3×6) generator matrix

$$G = \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 & 1 \end{bmatrix}$$

Assume that a 3-bit message is encoded using G and transmitted over a noisy channel, and that the received message is $r = (111110)$. Assume that at most one bit error occurs during the transmission of the message.

Compute the syndrome of r and use it to detect and correct any error in the transmission.